

Regime-Adaptive Risk-Aware MPPI Control for Force-Aware Robotic Wiping with Temporal-Difference World Models

Zhiheng Li¹

¹Independent Researcher, Shanghai, China

Corresponding author: Zhiheng Li (email: LLzhiheng@outlook.com).

ABSTRACT Force-aware wiping requires contact acquisition, force regulation, transient control, and surface progression. We introduce regime-adaptive risk-aware model-predictive path integral (MPPI) planning for direct temporal-difference model-predictive control (TD-MPC2) wiping. The method retains requested force in latent dynamics and adapts planning costs and computation from causal contact regimes and learned next-force predictions. Across 270 paired factor-separated evaluations, it raises compound pass from 85/135 to 100/135 (+11.11 percentage points; block- and seed-aware 95% interval [3.70, 18.52]), reduces mean relative error by 1.83 percentage points, and uses 8.0 fewer samples and 25.7 ms less median planning time. Task completion remains 132/135 versus 133/135, and both planners record two samples above 15 N. A 375-run one-factor panel identifies force bias, 50-ms delay, and registered-normal error as distinct sensitivities. In a zero-shot MuJoCo stress test, all 30 flat and inclined evaluations achieve the compound outcome, whereas none of 15 weakly curved evaluations does. Regime adaptation therefore improves the tracking–computation trade-off in simulation, while curved contact remains the principal portability boundary.

INDEX TERMS contact-rich manipulation, cross-simulator transfer, force-aware control, model-predictive path integral control, robotic wiping, TD-MPC2

I. Introduction

Robotic wiping is a contact-rich control problem in which geometric progress and force regulation must succeed together. The tool must first acquire the surface, approach a requested load without a large transient, maintain useful contact during motion, and recover when contact is lost. These phases place different demands on a controller. Contact acquisition rewards decisive inward motion, stable tracking rewards accurate force regulation, and rapid force growth calls for conservative action. A planning objective and sampling budget that remain fixed throughout an episode need not serve all three requirements equally well.

Temporal-difference model-predictive control (TD-MPC2) offers a natural basis for this problem: a learned actor provides a policy prior, while a latent world model evaluates action sequences online [1], [2]. Earlier ForceWipe experiments showed that behaviour-cloning (BC)-assisted TD-MPC2 could complete controlled first-pass wiping directly at 5, 8, and 12 N. They also exposed three limits. First, a hard measured-force threshold routed between the actor and planner without adapting the planning objective itself. Sec-

ond, the 12 N tier retained a systematic negative force bias. Third, the matched actor-only comparison did not resolve a general performance benefit from online planning despite a clear computation cost. These observations motivate a method that changes what the planner optimizes and how much it searches, rather than merely changing when planning is enabled.

We introduce regime-adaptive risk-aware MPPI (RA-RMPPI), a unified planner for direct TD-MPC2 wiping. The requested force is encoded by a dedicated condition branch and retained through imagined latent transitions. Learned next-force predictions provide a causal risk signal. Smooth causal memberships represent acquisition, approach, tracking, recovery, and transient risk. They continuously adjust contact, tracking, transient, and actor-anchor costs together with the MPPI sample count, optimization iterations, and exploration scale. The actor remains the proposal distribution in every regime; the historical hard actor–planner switch is removed.

The empirical design separates method selection from evaluation. Five unfiltered seeds share the same data and

update budget. A four-arm ablation isolates adaptive objectives and adaptive computation, while additional ablations examine behaviour-cloning strength, transient-risk representation, and the historical routing threshold. The method is then evaluated on new factor-separated shifts in surface geometry and contact dynamics, avoiding the path–surface–tool confounding of the earlier combined out-of-distribution (OOD) blocks. Across 270 paired evaluations, the complete method raises compound pass from 85/135 to 100/135 while using fewer planning samples and less host planning time. The gain is concentrated in force regulation: task completion and sampled force violations do not improve, and the cylindrical 12 N condition remains a failure boundary. A zero-shot MuJoCo stress test sharpens that boundary: all 30 flat and inclined evaluations achieve the compound outcome, whereas none of the 15 cylindrical evaluations does.

The paper makes three contributions. First, it presents a force-conditioned latent dynamics construction that preserves the requested force during multi-step imagination. Second, it develops a regime-adaptive risk objective and compute schedule that unifies acquisition, tracking, and recovery without a discrete actor–planner router. Third, it provides matched component and training ablations, including negative evidence for a more elaborate transient-envelope representation, together with factor-separated and cross-engine simulation evaluation. The resulting evidence identifies a statistically resolved tracking–computation improvement while preserving the method’s unresolved curvature and force-tail limits.

II. Related Work

A. Robot Wiping and Surface Interaction

Surface processing couples geometric coverage with sustained contact. Probabilistic Surface Interaction Primitives (ProSIP) conditions compliant motions on tool-centre-point (TCP) pose, surface geometry, and context [3]. Adaptive Wiping combines few-shot imitation, force–torque feedback, and pre-trained object representations [4]. Liu *et al.* study quality-critical wiping over randomised curvatures, frictions, and waypoints using a visual-language curriculum; their 20 Hz pose policy operates through an operational-space controller around a 60 N force target [5]. We examine direct learned Cartesian displacements at 5, 8, and 12 N.

Variable Impedance Control in End-Effector Space (VICES) uses variable impedance as a reinforcement-learning (RL) action space, including for wiping [6]. Beltran-Hernandez *et al.* combine RL with parallel position–force or admittance control and a real-robot fail-safe [7]. Zhu *et al.* couple an image-based equilibrium-point/impedance policy with faster contact-aware control [8]. Force-Guided Exploration for Robust Contact-Rich Manipulation under Uncertainty (FORGE) conditions assembly policies on a maximum allowable force and uses dynamics randomisation for hardware transfer [9]. These approaches structure learned contact through the action space or execution layer. Our

matched comparison uses a common Cartesian-delta interface for TD-MPC2 and proximal policy optimisation (PPO), with adaptive admittance as a separate reference.

Curved-surface methods also control tool orientation. Armesto *et al.* transfer planar wiping to unknown curves by estimating the contact constraint and aligning the tool from force/torque feedback [10]. Li *et al.* estimate pose deviation, lift a planar path onto an unknown workpiece, and close a normal-force loop [11]. Our registered frame supplies a local normal, but Eq. (2) controls translation only. Tool posture is not a policy action, a distinction relevant to the observed curved-contact boundary.

B. Force Control and Contact Simulation

Hybrid position–force control separates directional objectives [12]; impedance and admittance regulate complementary motion–force relations [13], [14]. Our adaptive admittance baseline shares the time-causal observation and actuator action box. Force-feedback model-predictive control (MPC) offers another approach: Kleff *et al.* incorporate measured force/torque through robot-actuation dynamics [15], while Jordana *et al.* estimate force-model discrepancy online without an explicit contact-force evolution model [16]. Our controller instead learns latent dynamics conditioned on scalar normal force and issues displacements without a separate force-control layer.

Runtime shields modify policy proposals [17], and control barrier functions can enforce forward-invariance conditions [18], [19]. Here, learned actions execute without an external shield or force-dependent post-processing. Sampled force exceedances are therefore empirical observations for a specified numerical configuration, not formal invariance guarantees. Contact models, relaxations, and solvers can produce different application-level behaviour [20]; Sec. IX tests the sensitivity of our force-tail claims.

C. Direct Model-Based Reinforcement Learning

Temporal-difference model-predictive control (TD-MPC) and TD-MPC2 combine latent dynamics, value prediction, and online planning [1], [21]. We retain actor-guided MPPI, replacing the earlier force-threshold router with continuous causal memberships that adapt planning costs and search budget. A same-checkpoint fixed-planner ablation isolates this adaptation from the learned actor and world model. PPO [22] provides a direct model-free comparator, with its training budget and hyperparameters examined separately.

Structured alternatives include residual RL around a conventional controller [23], parameterised action primitives and Manipulation Primitive-augmented Reinforcement Learning (MAPLE) [24], [25], and impedance primitive-augmented hierarchical reinforcement learning (IMP-HRL) [26]. Our study instead tests regime-adaptive planning within direct actor/world-model control, using controlled variation, weak-curvature transfer, and claim-specific numerical auditing.

III. Simulation Task and Evaluation Suite

We use ForceWipe as a controlled simulation testbed for direct first-pass wiping. A successful evaluation must traverse the path, maintain useful contact, track the target normal force, and avoid recorded force-limit exceedances. The test cases vary path geometry, surface orientation, tool profile, contact parameters, disturbances, and weak surface curvature. The resulting suites define controlled and transfer conditions for the same direct action interface.

A. Direct First-Pass Task

A seven-degree-of-freedom Franka Emika Panda manipulates a wiping tool in ManiSkill3 with the PhysX central processing unit (CPU) backend [27], [28]. Physics and control advance together at 100 Hz. At each step, the local task frame is formed by the path tangent $\mathbf{t}_t$, outward surface normal $\mathbf{n}_t$, and cross-track axis $\mathbf{c}_t = \mathbf{n}_t \times \mathbf{t}_t$. The controller outputs

$$\mathbf{a}_t = \begin{bmatrix} a_t^{\parallel} & a_t^{\perp} & a_t^n \end{bmatrix}^T \in [-1, 1]^3, \quad (1)$$

where the components request along-path, cross-track, and inward-normal displacements. The fixed, force-independent actuator map is

$$\Delta \mathbf{x}_t = (0.50 \text{ mm}) a_t^{\parallel} \mathbf{t}_t + (0.25 \text{ mm}) a_t^{\perp} \mathbf{c}_t - (0.50 \text{ mm}) a_t^n \mathbf{n}_t. \quad (2)$$

Post-policy processing consists exclusively of action-box clipping and a force-independent coordinate transform in (2).

The tangent and normal are evaluated from the registered nominal path and surface model at the simulator-state tool pose. They are not inferred from a camera, force sensor, or the TD-MPC2 latent state. Consequently, “direct” in this study denotes the absence of a separate force-control or shielding layer; it does not denote model-free geometric perception. This registered-frame assumption is shared by all compared methods.

At decision step t , F_t denotes the force supplied to the causal observation before $\mathbf{a}_t$ is selected. We reserve F_t^{aud} for the native force read back after the corresponding physics step and attached to that transition for evaluation. The target normal force is $F^* \in \{5, 8, 12\}$ N. The online contact indicator uses $F_t \geq F_{\min} = 3$ N, whereas evaluation metrics use the post-step contact set $F_t^{\text{aud}} \geq F_{\min}$ defined in Sec. D. A strict sampled violation is $F_t^{\text{aud}} > F_{\max} = 15$ N. The statement “zero violations” therefore means that no recorded 100-Hz sample exceeded 15 N. All force-limit results use this native sampled definition. The three force targets, the 3-N contact threshold, and the 15-N sampled limit form the frozen operating contract of this simulation task. They are used identically by every method and were fixed before the controlled and factor-separated evaluations; they are not presented as general robot or human-contact safety standards.

Each path is divided into 20 normalized progress bins. A bin is marked complete once it has accumulated at least two samples satisfying $F_{\min} \leq F_t^{\text{aud}} \leq F_{\max}$, corresponding to

20 ms at the simulator rate. A task pass requires a single first-pass lifecycle to terminate with all 20 bins complete and progress $s \geq 0.98$. Re-wiping is disabled in all three evaluation suites. The dose endpoint is synthetic contact coverage and serves as the completion proxy.

B. Controlled Evaluation Blocks

The initial and held-out controlled suites use three surface orientations (flat, 5° incline, and 10° incline), five path families (line, diagonal, arc, S-curve, and polyline), and three tool profiles (wide-soft, narrow-soft, and wide-hard). Their synthetic residual fields comprise single-blob, islands, edge-dense, stripe or multimodal, and Gaussian-random-field patterns. These factors are assembled into prespecified blocks rather than sampled as an independent factorial design.

The five initial blocks Q0–Q4 use the nominal contact plant and no exogenous disturbance. They vary path length between approximately 0.156 and 0.166 m while changing geometry, tool, and residual field by block. The ten previously unused held-out blocks C0–C9 form two disjoint groups. C0–C4 repeat the five path families on a nominal plant. C5–C9 pair the same roster positions with randomised path length, friction, support stiffness, effective mass, damping, and restitution. Their frozen definitions also label designated blocks with lateral or normal impulses, reference noise, and delayed sensor noise. A later binding audit established the physical impulse call chain but did not establish that the declared reference-noise and sensor-noise signals reached the direct controller’s learner-visible force path. We therefore make no robustness claim for those two declared noise conditions. Because several verified attributes change jointly, grouped path, surface, tool, residual, and plant summaries quantify block-level transfer rather than single-factor effects.

C. Weak-Curvature Out-of-Distribution Blocks

The matched OOD study contains six previously unused, prespecified blocks F0–F5. All surfaces are static cylinders with radii 1.2–1.8 m. Its paths are arc, S-curve, or polyline trajectories with lengths 0.13–0.205 m. Across the six blocks, the study includes all three tool profiles, six residual families, randomised friction and contact plant parameters, and predefined disturbance labels. Physical lateral and normal impulse call chains are present in their designated blocks; the declared reference-noise and sensor-noise/latency effects are excluded from the evidence claims because their runtime binding was not established. Table 1 summarizes the evaluation stages.

The controlled and weak-curvature suites are reported separately to preserve their deliberately different scenario distributions.

D. Evaluation Metrics and Success Criteria

Force tracking is computed over the finite contact set

$$\mathcal{T}_c = \{t \mid F_t^{\text{aud}} \geq 3 \text{ N}\}. \quad (3)$$

TABLE 1. Prespecified simulation suites. Evaluation totals are Cartesian products of blocks, force targets, and method replicates; blocks are the environment units. For the deployment-gap panel, 375+45 denotes new stress evaluations plus reused nominal evaluations.

Stage	Blocks	Targets	Replicates	Evaluations	Main variation
Controlled evaluation	5	3	5 checkpoints	75	Five paths; flat/5°/10° surfaces; three tools; nominal plant
Held-out controlled evaluation	10	3	5 checkpoints	150	Five nominal multipath blocks plus five randomised-plant/disturbance blocks
Matched weak-curvature OOD	6	3	method dependent	288	Static weak-curvature cylinders, three paths, three tools, randomised plant, and disturbances
Numerical sentinel panel	4	1	one checkpoint	36	Four 12 N OOD baseline reproductions plus 32 solver, contact-offset, mesh, and interaction cells
Deployment-gap stress	3	3	5 checkpoints	375+45	Learner-visible force noise/bias, delay, TCP-position error, and registered-frame normal error

If $\mathcal{T}_c$ is empty, the force-tracking metrics are undefined and the evaluation fails the tracking criterion. For a nonempty contact set, we report mean relative error (MRE), root-mean-square error (RMSE), and normalized root-mean-square error (NRMSE):

$$\begin{aligned}
\bar{F}_c &= \frac{1}{|\mathcal{T}_c|} \sum_{t \in \mathcal{T}_c} F_t^{\text{aud}}, \\
b_F &= \bar{F}_c - F^*, \quad \text{MRE} = \frac{|b_F|}{F^*}, \\
\text{RMSE} &= \sqrt{\frac{1}{|\mathcal{T}_c|} \sum_{t \in \mathcal{T}_c} (F_t^{\text{aud}} - F^*)^2}, \\
\text{NRMSE} &= \frac{\text{RMSE}}{F^*}. \tag{4}
\end{aligned}$$

Tracking is considered successful when $\text{MRE} \leq 0.15$ and $\text{NRMSE} \leq 0.20$. Signed bias is retained so systematic under-tracking remains visible even when both tolerances are met. These two tolerances are likewise study-defined acceptance criteria frozen with the evaluation protocol, rather than thresholds inferred from the reported trials.

The force evaluation records episode peak force, peak overshoot $\max(\max_t F_t^{\text{aud}} - F^*, 0)$, peak-force mean, 95th percentile (P95), and maximum, and sample counts above 14.5 and 15 N. The 14.5 N count is descriptive headroom; the sampled-force-limit criterion requires zero samples above 15 N. Contact metrics include contact fraction and, after first contact, the number, total duration, and maximum duration of maximal consecutive intervals below 3 N. Force-tail quantities describe the frozen numerical configuration in which they were observed. Section IX separately tests their sensitivity to selected simulator settings.

An evaluation satisfies deployment integrity when every trace confirms direct-action interface compliance: the selected learned branch issued the executed action, followed by the fixed map in (2). A *compound pass* is the intersection of task completion, tracking, sampled-force-limit compliance, and deployment integrity. Nonfinite force or an empty contact set counts as an evaluation failure. Planning time and

the actor/MPPI action fractions are descriptive computational metrics.

E. Scope of the Suite

The blocks are fixed conditions, with each block defining an environment unit. Replicated checkpoint–block–target trials increase the precision of within-design comparisons. The suite characterizes direct force-aware wiping under controlled variation and weak-curvature transfer; Sec. X defines the corresponding deployment scope.

IV. Regime-Adaptive Risk-Aware World-Model Control

The proposed controller retains the learned latent dynamics, force predictor, and policy prior of TD-MPC2. Its contribution is to couple these components through a causal, state-dependent planning objective rather than a fixed force threshold. The policy prior proposes candidate actions in every contact regime. A single model-predictive path integral (MPPI) planner then changes its objective and computational budget according to the current force state and the model’s predicted transient-risk estimate. No classical force controller, post-hoc action projection, or evaluation-time teacher is introduced. Figure 1 summarizes the resulting data flow.

Let F_t , F^* , and $\dot{F}_t$ denote measured force, requested force, and measured force rate. Unlike an observation-only target input, the requested force is encoded by a dedicated condition branch $\mathbf{c}_t = e_\theta(F^*)$. The latent state concatenates a base state and this condition, $\mathbf{z}_t = [h_\theta(\mathbf{o}_t); \mathbf{c}_t]$, and the imagined transition is

$$\mathbf{z}_{t+1} = [f_\theta(\mathbf{z}_t, \mathbf{a}_t, \mathbf{c}_t); \mathbf{c}_t]. \tag{5}$$

The copied condition prevents the requested force from being diluted over a multi-step rollout while the learned base dynamics remain action dependent. For candidate action $\mathbf{a}_t$, the selected world-model risk representation is the predicted next force

$$U_{t+1} = \hat{F}_{t+1}. \tag{6}$$

During an H -step imagined rollout, this head is evaluated after every latent transition, producing the causal transient

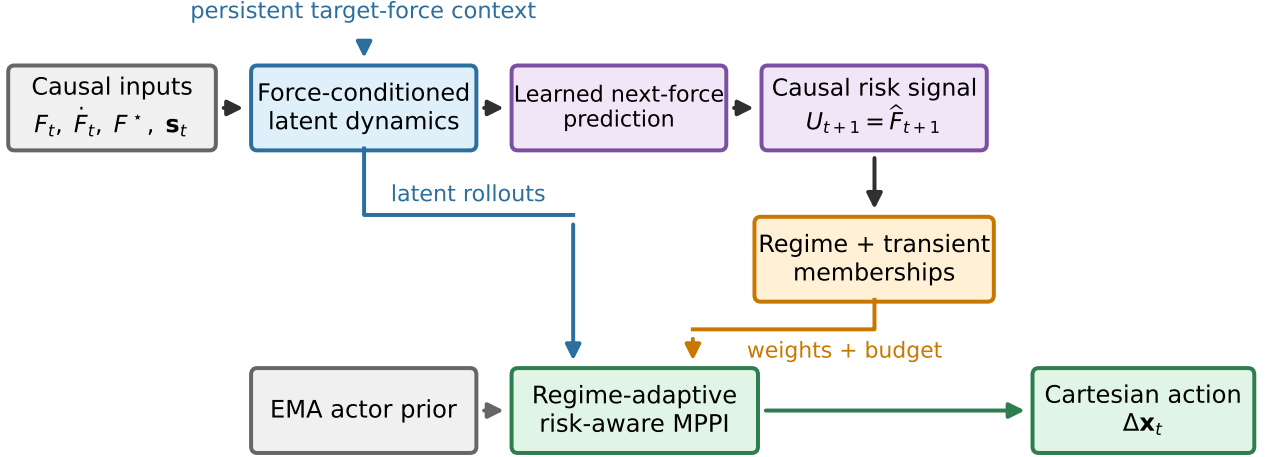


FIGURE 1. Force-conditioned regime-adaptive risk-aware MPPI architecture.

forecast $\hat{\mathbf{F}}_{t+1:t+H}$ used by the tracking and transient costs. This choice follows a matched development ablation against a learned transient-envelope point prediction and a target-calibrated alternative; neither alternative improved the pre-specified outcomes.

Smooth causal memberships

$$\boldsymbol{\pi}_t = [\pi_t^{\text{acq}}, \pi_t^{\text{app}}, \pi_t^{\text{track}}, \pi_t^{\text{rec}}] \quad (7)$$

distinguish initial contact acquisition, approach to the force target, stable tracking, and recovery after contact loss. A fifth, overlapping membership π_t^{tr} measures transient risk from U_{t+1} and positive force rate. These memberships depend only on $(F_t, F^*, \dot{F}_t, U_{t+1})$ and episode-level contact history. They replace the previous hard switch at $0.85F^*$.

Let $r_t = \max(F_t, 0)/F^*$, let $\sigma(x) = (1 + \exp(-x))^{-1}$, and let $h_t \in \{0, 1\}$ record whether contact has occurred earlier in the episode. The unnormalized phase scores are

$$\tilde{\pi}_t^{\text{acq}} = (1 - h_t)\sigma\left(10\frac{3 - F_t}{F^*}\right), \quad (8)$$

$$\tilde{\pi}_t^{\text{rec}} = h_t\sigma(10(0.65 - r_t)), \quad (9)$$

$$\tilde{\pi}_t^{\text{app}} = \sigma(10(0.90 - r_t)), \quad (10)$$

$$\tilde{\pi}_t^{\text{track}} = \exp\left[-\frac{1}{2}\left(\frac{r_t - 1}{0.15}\right)^2\right]. \quad (11)$$

Normalizing these four scores gives $\boldsymbol{\pi}_t$. Transient membership overlaps the phase simplex:

$$\begin{aligned} \pi_t^{\text{tr}} = 1 - & \left[1 - \sigma\left(10\frac{U_{t+1} - 14.5}{0.5}\right)\right] \\ & \times \left[1 - \sigma\left(10\left(\frac{\max(\dot{F}_t, 0)}{80} - 1\right)\right)\right]. \end{aligned} \quad (12)$$

Forces are in newtons and force rate is in newtons per second. The membership is therefore high when either the predicted next force approaches the force limit or measured force rises rapidly.

The force thresholds in these memberships inherit the task contract in Sec. A. The remaining ratios, bandwidth, sigmoid gain, and force-rate scale are development-fixed design parameters: they create smooth, overlapping transitions rather than learned decision boundaries. They were held fixed before the nine-block evaluation, and no coefficient in Eqs. (12)–(16) was fitted on those final blocks.

For a candidate sequence $\mathbf{a}_{t:t+H-1}$, the planner maximizes

$$\begin{aligned} J_t = J_{\text{task}} - \lambda_c(\boldsymbol{\pi}_t)J_{\text{contact}} - \lambda_f(\boldsymbol{\pi}_t)J_{\text{tracking}} \\ - \lambda_r(\boldsymbol{\pi}_t)J_{\text{transient}} - \lambda_a(\boldsymbol{\pi}_t)J_{\text{anchor}}, \end{aligned} \quad (13)$$

where J_{tracking} penalizes predicted force error, J_{contact} penalizes predicted loss of useful contact, $J_{\text{transient}}$ penalizes predicted next force above the soft force margin and rejects candidates at the sampled force limit, and J_{anchor} regularizes search around the learned policy prior. The four coefficients are continuous functions of the causal memberships. Contact restoration is emphasized during acquisition and recovery, force accuracy during approach and tracking, and transient restraint when predicted risk or positive force rate rises.

Specifically, with $q_t = \text{clip}_{[0,1]}(\pi_t^{\text{tr}} + 0.35\pi_t^{\text{rec}} + 0.15\pi_t^{\text{acq}})$, the implemented weights are

$$\lambda_c = 8(\pi_t^{\text{acq}} + \pi_t^{\text{rec}} + 0.25\pi_t^{\text{app}}), \quad (14)$$

$$\lambda_f = 6(0.20 + 0.80\pi_t^{\text{track}} + 0.45\pi_t^{\text{app}})(1 - 0.70\pi_t^{\text{tr}}), \quad (15)$$

$$\lambda_r = 100(0.10 + 1.90q_t), \quad \lambda_a = 1 + 1.50q_t. \quad (16)$$

The same memberships determine the MPPI sample count, number of optimization iterations, and proposal standard deviation. Stable tracking uses the smallest budget. Contact acquisition and recovery use a broader proposal distribution, whereas a high-risk transient uses more samples with a narrower distribution. The priority is transient ($\pi_t^{\text{tr}} \geq 0.5$), recovery ($\pi_t^{\text{rec}} \geq \max(\pi_t^{\text{acq}}, 0.35)$), acquisition ($\pi_t^{\text{acq}} \geq 0.35$), approach ($\pi_t^{\text{app}} > \pi_t^{\text{track}}$), and then tracking. This changes computation and exploration without changing the Cartesian action interface. Every control step records the memberships,

TABLE 2. Regime-dependent MPPI budget.

Regime	Samples	Elites	Iterations	Max. std.
Transient	256	32	4	0.10
Recovery	160	20	3	0.20
Acquisition	160	20	3	0.20
Approach	128	16	3	0.16
Tracking	96	12	2	0.12

objective weights, next-force risk prediction, selected budget, and issued action so that the adaptive mechanism can be evaluated separately from task outcome.

The fixed-planner anchor uses 128 samples, 16 elites, four optimization iterations, and maximum proposal standard deviation 0.25. The adaptive table keeps the elite fraction at $1/8$, bounds sampling between $0.75\times$ and $2\times$ the 128-sample anchor, and uses two to four optimization iterations. These choices encode a monotone compute allocation across nominal tracking, contact restoration, and transient risk; the matched component ablation tests the allocation as a whole rather than selecting its entries on final outcomes.

The component study compares this complete formulation with fixed-objective, adaptive-objective-only, and adaptive-compute-only variants before the factor-separated evaluation.

V. Experimental Design

The experiments address five questions. **RQ1** asks whether adapting the planning objective and search budget improves direct wiping relative to a fixed planner built from the same checkpoint. **RQ2** separates the contributions of objective adaptation, compute adaptation, and risk representation. **RQ3** examines how behaviour-cloning (BC) strength affects the actor proposal and closed-loop contact. **RQ4** tests whether the selected method transfers under isolated surface, path, friction, and support-stiffness shifts. **RQ5** asks how the frozen method responds to learner-visible sensing, latency, and registered-frame errors, and whether an independently implemented simulator is sufficiently task-equivalent for matched replication. It then asks which task capabilities persist when the frozen policy is deployed zero-shot despite a documented residual engine gap.

A. Training and Seed Policy

All learned arms use five predetermined seeds. Each seed receives 82,774 training transitions and 4,000 optimization updates. The update schedule contains a target-conditioning gate, BC pretraining, and joint world-model, value, force-prediction, envelope-prediction, actor, and BC optimization. No seed is removed on the basis of training or evaluation performance. The envelope output remains available only for the development mechanism ablation; the selected controller uses the learned next-force prediction. The transition count is the complete frozen source dataset used by each TD-MPC2

seed; equal data and update budgets are maintained across its training ablations. Seeds are experimental replicates, and none is screened or replaced.

The BC study retrains the same five-seed roster with coefficients 0, 0.5, and 2. The architecture, source transitions, update count, target-force roster, and planner are unchanged. The selected coefficient is the smallest tested setting that maintains contact, completion, and sampled-limit compliance for all seeds on the development screen. The three-point grid represents absent, intermediate, and strong proposal anchoring; it is a bounded mechanism study, not a claim that 2 is a universal optimum.

B. Matched Planner Ablations

Four planner arms share each force-conditioned checkpoint. M1 uses fixed objective weights and a fixed MPPI budget. M2 adapts the objective only. M3 adapts sample count, optimization iterations, and exploration scale only. M4 combines both mechanisms. The development screen crosses the four arms with five seeds and three force targets under common scenario and per-step random streams.

A second three-arm ablation changes the risk representation inside M4. The *force-only* arm uses predicted next force, the *point-envelope* arm additionally uses the predicted transient envelope, and the *calibrated-envelope* arm adds the target-specific validation residual. This comparison tests whether the uncertainty allowance contributes beyond the point predictor. The mode is selected lexicographically by sampled force violations, task success, strict tracking passes, mean NRMSE, and mean MRE.

The historical controller used a hard actor-MPPI switch at $F_t = 0.85F^*$. We evaluate activation fractions from 0.70 to 0.95 using the frozen historical checkpoints, shared scenarios, and shared step-level random streams. This study measures the sensitivity of the earlier router; RA-RMPPI itself contains no hard routing threshold.

C. Factor-Separated Evaluation

After method selection, M1 and M4 are evaluated on nine new fixed blocks. A reference block uses a flat surface, straight path, friction coefficient 0.5, and support stiffness 2500 N/m. Two surface blocks replace the surface with a 10-degree incline or a 1.2-m-radius cylinder. Two path blocks use an arc or S-curve. Two friction blocks use coefficients 0.3 or 0.7, and two support blocks use 1600 or 3400 N/m. Every shifted block changes only its declared factor relative to the reference; tool, residual field, disturbance, path length, mass, damping ratio, and restitution remain fixed. The friction values bracket the reference by ± 0.2 , and the stiffness values bracket it by ± 900 N/m. Together with the two fixed surface and path contrasts, they are prespecified stress levels for factor isolation, not samples from a material or deployment population.

The full design contains $9 \times 3 \times 5 \times 2 = 270$ evaluations. M1 and M4 share the scenario and per-control-step random

stream within every seed–block–target cell. The primary outcome is a compound pass requiring task completion, tracking mean relative error (MRE) at most 15%, target-normalized root mean square error (NRMSE) at most 20%, zero native samples above 15 N, and deployment integrity. Continuous force bias, MRE, NRMSE, contact loss, sampled peak force, planning latency, and planning sample count are reported separately. Paired intervals resample both block and training seed. Factor panels are analysed separately so that a surface effect is not attributed to simultaneous changes in path, tool, or plant parameters. The compound pass is the sole primary outcome. Other intervals quantify effect size and within-design stability and are not treated as additional multiplicity-adjusted confirmatory tests. All 270 rows in the Cartesian product are retained, and inference is restricted to the nine fixed blocks and five checkpoint seeds represented by the resampling scheme.

D. Comparator Evidence

Direct PPO uses the same deployment observation and action interface. A bounded follow-up tests whether its original result is specific to the 82,774-transition budget or a single default configuration. For each of five predetermined seeds, the default setting is checkpointed after $1\times$, $2\times$, and $4\times$ that budget. Six one-factor variants change the learning rate, clipping ratio, or entropy coefficient at $2\times$ budget. All nine endpoints are evaluated on six development blocks. A prespecified lexicographic rule selects one endpoint, which is then evaluated without further tuning on the matched weak-curvature and factor-separated suites. PPO is trained online whereas TD-MPC2 uses a frozen source dataset, so the comparison matches simulator transitions rather than optimizer updates. Adaptive admittance is reported target by target as a deterministic structured-control reference.

E. Deployment-Gap and Cross-Simulator Probes

The deployment-gap study freezes the selected controller and varies one learner-visible factor at a time. The factors are force white noise at 0.035 N root mean square, episode force biases from -0.8 to 0.8 N, observation delays of 10, 20, and 50 ms, TCP-position errors of 0.05 and 0.10 mm, and registered-frame normal errors of 1, 3, and 5 degrees. A reference block traces the complete level curves; two additional fixed blocks repeat the most severe level of each factor. Force perturbations affect the policy observation but not the native-force outcome metric, while the normal error is applied consistently to geometric observations and task-frame action mapping. The analysis reports paired degradation curves over five frozen checkpoints and three force targets. These levels are controlled simulation stresses rather than estimates of a joint hardware-error distribution. The force-noise, force-accuracy, and position-repeatability scales follow the manufacturer handbook [29]; the latency and frame-error levels are study-defined stresses.

A separate policy-independent gate measures how closely a MuJoCo task port reproduces the frozen PhysX task. Fixed open-loop actuator and contact motions are divided into development and held-out sets. The fixed tolerances are force NRMSE at most 20%, relative-position RMSE at most 0.5 mm, contact-onset error at most five 100-Hz samples, and peak-force error at most 20%. The revised port meets the free-space actuator and held-out normal-cycle criteria but not the bidirectional tangential-position criterion. It is therefore not used as a matched solver replication.

We nevertheless freeze a separate zero-shot stress test to measure portability under that documented mismatch. The five M4 checkpoints are deployed without retraining or post-result port tuning on the reference flat surface, the 10° incline, and the 1.2-m-radius cylinder at 5, 8, and 12 N. This gives $5 \times 3 \times 3 = 45$ evaluations with keys matched to the completed PhysX study. Task, tracking, sampled-limit safety, and their compound are reported by surface block. The test quantifies transfer to this MuJoCo port rather than hardware transfer or general simulator invariance.

VI. Method Development Results

All four planner arms completed all 60 development evaluations, with no sample above 15 N. Table 3 shows that adaptive objective and compute each recover part of the fixed-planner gap, while their combination gives the lowest mean relative error (MRE) and normalised root-mean-square error (NRMSE). The complete method reduces 12 N RMSE from 2.927 to 2.600 N and raises mean force from 9.308 to 9.714 N, although the strict 12 N tracking gate remains 0/5 for every arm.

A behaviour-cloning (BC) coefficient of 2 completes all 15 development runs without a sampled violation; removing BC prevents contact in all 15, whereas coefficient 0.5 yields 12/15 task passes and two samples above 15 N. Router thresholds change planner use smoothly rather than revealing a discrete operating point. The stable MPPI segment contains most of the 12 N negative bias, and alternate risk representations do not improve the frozen force-only representation. Complete development tables and sample-level decompositions are provided in the Supplementary Material.

VII. Factor-Separated Evaluation Results

All 270 scheduled evaluations completed. Table 4 summarizes the two 135-run arms. The adaptive method increases compound pass from 85 to 100 and strict tracking pass by the same amount. The paired difference is 11.11 percentage points with a block- and seed-aware 95% bootstrap interval of [3.70, 18.52]. Mean MRE and NRMSE decrease by 1.83 and 1.28 percentage points, respectively; both intervals exclude zero. Mean planning samples decrease by 7.97 and mean median planning time by 25.65 ms.

The benefit is a tracking–computation trade-off rather than a general safety gain. Task success changes by -0.74

TABLE 3. Matched planner-component development ablation. MRE and NRMSE are averaged over 15 seed–target evaluations per arm.

Arm	MRE (%)	NRMSE (%)	Samples	Median time (ms)
M1: fixed objective, fixed compute	12.03	14.66	128.0	112.18
M2: adaptive objective	10.94	13.77	128.0	107.86
M3: adaptive compute	10.26	13.52	121.2	80.58
M4: complete method	9.65	12.94	120.4	83.25

TABLE 4. Factor-separated evaluation over 135 runs per planner.

Outcome	Fixed	Adaptive
Task success	133	132
Tracking pass	85	100
Compound pass	85	100
Mean MRE (%)	12.16	10.33
Mean NRMSE (%)	15.07	13.78
Samples > 15 N	2	2
Maximum peak (N)	17.399	17.905
Planning samples	128.0	120.0
Median time (ms)	101.09	75.43

percentage points, interval $[-4.44, 0.00]$, and sampled-limit violations remain two in each arm. Mean peak force increases by 0.247 N, interval $[0.045, 0.428]$ N. At 12 N, tracking improves from 5/45 to 17/45, but task success changes from 43/45 to 42/45.

Figure 3 shows that the aggregate improvement is not uniform. Compound pass rises for the incline, both paths, both friction levels, both support stiffnesses, and the reference block. On the cylinder it falls from 7/15 to 6/15. The cylinder also contains every task failure and sampled force-limit violation: its maximum peak is 17.399 N for the fixed planner and 17.905 N for the adaptive planner. The method therefore improves regulation under most isolated shifts but does not resolve weak curvature combined with a 12 N target.

For comparator context, an earlier matched simulation suite used the same Cartesian action interface. The original direct PPO configuration achieved 0/90 task completions, while the fixed TD-MPC2 planner and actor-only control achieved 19/90 and 18/90 compound passes. Adaptive admittance achieved 11/18 compound passes and was strongest at 5 and 8 N but degraded at 12 N. The admittance values are one deterministic run per cell and do not enter the paired interval analysis. For the fixed-planner versus actor-only comparison, task and tracking differences were 2.22 and 3.33 percentage points, with intervals $[-6.67, 13.33]$ and $[-4.44, 13.33]$. The design therefore resolved neither a benefit nor equivalence for the historical online planner. These results establish learned and structured-control references; the paired M1–M4 experiment above isolates the contribution of regime adaptation.

The expanded PPO study covers three budgets and six one-factor variants over five seeds. The development rule selected clip ratio 0.1 at 165,548 transitions per seed, with 3/90 compound and 11/90 task passes. On the two confirmation suites, that endpoint achieved 0/90 task and compound passes on the matched weak-curvature set, followed by 19/135 task and 6/135 compound passes on the factor-separated set. All six compound passes came from one seed. By comparison, RA-RMPPI achieved 100/135 compound passes on the same factor-separated cells. This cross-family contrast is descriptive because the checkpoint seeds are independent; it is not included in the paired M1–M4 interval. The Supplementary Material reports every PPO endpoint. The result shows that the weak PPO reference persisted across the tested budget and local hyperparameter range, without making an algorithm-independent claim about PPO.

The original PPO training record provides the matched-budget anchor. Across 645 training episodes, 3 completed the task and 470 contained a sampled force-limit violation. Figure 2 shows that training reduced the violation rate but did not produce reliable task completion at 82,774 transitions per seed.

VIII. Deployment-Gap and Cross-Engine Results

The frozen M4 controller was evaluated in 375 new one-factor stress evaluations. Together with 45 reused nominal evaluations, the panel contains 420 observations over the same three blocks, five checkpoints, and three force targets. Every stress trace retained direct TD-MPC2 action authority; the native simulator force remained the outcome signal while only the learner-visible observation or registered task frame was perturbed.

The signed force bias produced the clearest regulation response. A +0.8 N bias reduced compound pass from 24/45 under nominal observation to 12/45, including 11/15 to 3/15 in B0 and 7/15 to 0/15 in S1. The corresponding paired changes were -53.33 percentage points in B0 (95% pointwise interval $[-66.67, -40.00]$) and -46.67 points in S1 ($[-60.00, -33.33]$). A -0.8 N bias instead yielded 31/45 compound passes, but it also reduced task and sampled-limit safety to 40/45 and raised the maximum native-force peak to 22.337 N. The signed intervention therefore exposes observation bias rather than a uniformly beneficial operating offset.

Delay and frame error affected different parts of the task. At 50 ms, task and sampled-limit safety fell from 42/45

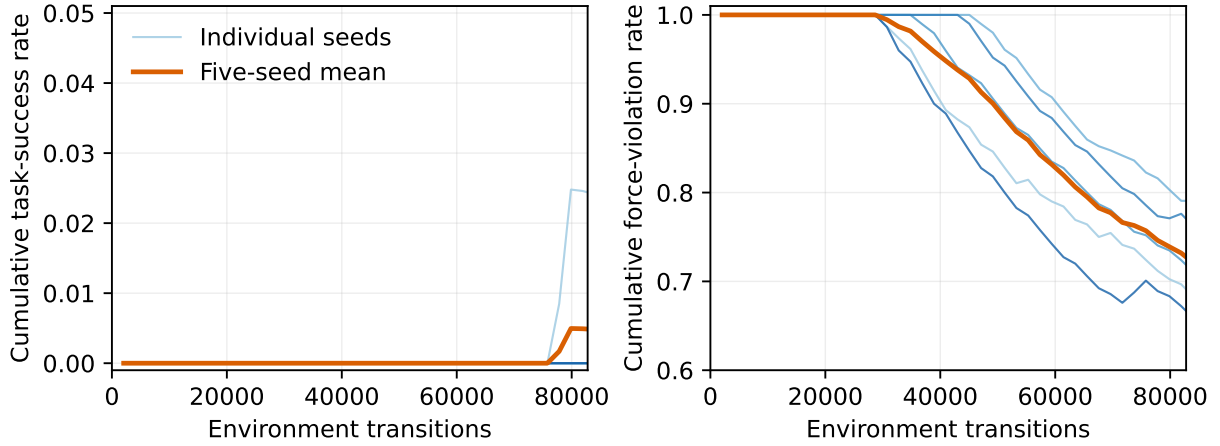


FIGURE 2. Original matched-budget direct PPO training over five seeds and 82,774 environment transitions per seed.

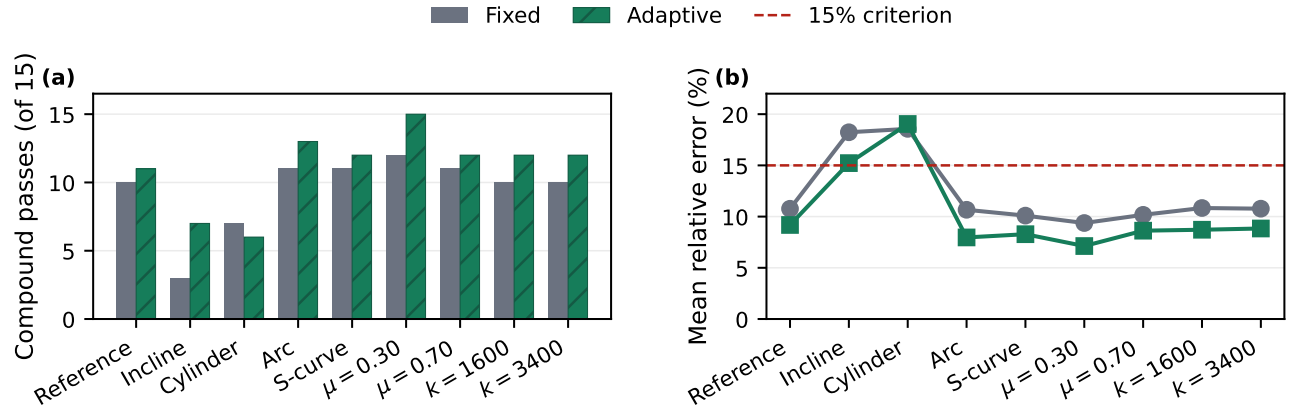


FIGURE 3. Factor-separated task-and-tracking outcome and mean force error; 15 evaluations per factor level and planner.

TABLE 5. Severe deployment-gap endpoints. Compound counts are per 15 evaluations in each fixed block; task, tracking, and sampled-limit safety counts are over 45 evaluations. Peak force uses the native simulator signal.

Condition	B0 comp.	S1 comp.	S2 comp.	Task	Tracking	Safety	Max peak (N)
Nominal	11/15	7/15	6/15	42/45	24/45	43/45	17.905
Force noise 0.035 N RMS	12/15	6/15	7/15	41/45	25/45	41/45	18.679
Force bias -0.8 N	14/15	13/15	4/15	40/45	31/45	40/45	22.337
Force bias $+0.8$ N	3/15	0/15	9/15	42/45	12/45	43/45	18.031
Delay 50 ms	11/15	12/15	5/15	34/45	28/45	35/45	26.084
TCP error 0.10 mm	11/15	7/15	7/15	44/45	25/45	44/45	18.491
Normal error 5°	9/15	8/15	5/15	30/45	24/45	41/45	21.316

and 43/45 to 34/45 and 35/45, and the maximum peak reached 26.084 N. A 5-degree registered-normal error reduced task completion to 30/45 and compound pass to 22/45. By contrast, 0.10-mm TCP-position error produced 25/45 compound passes versus 24/45 nominal, and 0.035-N RMS force noise produced 25/45. These one-factor results identify signed force calibration, latency, and frame alignment as the main deployment sensitivities in the evaluated simulation blocks; their mixed effects across B0, S1, and S2 also rule out a single monotone robustness score.

A. Zero-Shot MuJoCo Stress Test

The independently implemented MuJoCo port was evaluated only after its open-loop mismatch had been recorded. Table 6 reports the 45 zero-shot evaluations and their matched PhysX cells. The five frozen checkpoints achieve task completion, tracking, and sampled-limit safety in all 30 flat and inclined evaluations. In the cylindrical block, however, all 15 evaluations fail each component: no episode completes, every trace exceeds 15 N, and the maximum peak reaches 20.24 N. The matched PhysX block is also the weakest of its three surfaces, but it retains 12/15 task successes,

6/15 tracking and compound passes, and 13/15 sampled-limit safety passes.

The engine change therefore preserves the complete measured outcome on the two planar geometries and breaks it consistently on weak curvature. Because the calibration gate retains a bidirectional tangential-position mismatch, this contrast is a portability stress result rather than a solver-equivalence claim. It shows that the flat and inclined capability is not confined to one engine implementation, while curved contact remains dependent on the policy, geometry mapping, and contact solver as a coupled system.

IX. Provenance and Numerical Audit

An independent audit re-read the 288 matched out-of-distribution traces, verified their hashes, and recomputed all 7,776 per-evaluation metric values without importing the experimental runner or finalizer. Every value and trace identity matched. This verifies the reported metrics, while using the same recorded trajectories.

The controller is time-causal but not sensor-complete. Normal force, tool pose, and velocity come from simulator state; a registered nominal surface model supplies the local normal and tangent used by both observation and Cartesian action mapping. A separate binding audit verified the designated lateral and normal impulses, but did not establish that four declared reference- or sensor-noise labels reached the learner-visible force path. Those labels are therefore excluded from noise-robustness claims.

The prespecified 12 N numerical panel reproduced four OOD sentinels and tested 32 paired interventions in solver iterations, contact offset, source mesh, and their interaction. The material peak/threshold rule triggered in 11/32 cells; task, tracking, and sampled-limit status changed in three, two, and two cells, respectively. Ten peak responses were reductions, whereas one increase crossed 15 N. Force-tail observations consequently describe the recorded numerical configuration rather than a general safety guarantee.

All requested source arc counts reached the mesh factory, but PhysX convex cooking returned baseline-equivalent collision payloads in 11/12 pure mesh cells. Only one 96-segment request changed cooked topology and triggered a material response. The arm therefore audits request-to-collision mapping, not multilevel mesh convergence. Detailed provenance, claim boundaries, and all paired numerical cells are reported in the Supplementary Material.

X. Discussion and Limitations

The factor-separated evaluation supports the proposed mechanism rather than a simple increase in planning effort. Adaptive computation uses fewer samples, adaptive objectives reduce force error, and their combination raises compound pass by 11.11 percentage points with an interval that excludes zero. The improvement is concentrated in tracking and is especially visible at 12 N, where strict tracking rises from 5/45 to 17/45. It does not increase task completion

or reduce sampled-limit violations. High-force regulation on the cylinder therefore remains the central performance limit.

The mechanism ablations also delimit the contribution. Moving the historical routing threshold changes how often MPPI is used but does not expose a natural switching point. Conversely, adding a transient-envelope head or its target-specific calibration does not improve the frozen development outcomes. The supported method is consequently the smaller combination of persistent force conditioning, learned next-force risk, continuous regime-conditioned costs, and adaptive compute allocation.

The BC ablation clarifies what *direct* control requires in this study. The evaluation controller contains no teacher or separate force regulator, but the actor proposal is learned with behavioural supervision. Removing that supervision leaves one-step force-model accuracy unchanged and nevertheless eliminates contact in every development run. The result attributes BC’s main role to proposal quality for contact acquisition and recovery, not to better force prediction. The moderate coefficient further shows that low offline action error is insufficient: its seed-dependent failures appear only in the closed loop.

Planning latency remains a deployment constraint. Dynamic compute reduces the mean sample count and host planning time in the development screen, but the simulation still runs at a 100-Hz control discretization rather than a demonstrated 100-Hz wall-clock planner. The method should therefore be read as a control and allocation mechanism, not as evidence of real-time execution on the present host.

The deployment-gap panel separates several practical sensitivities. A 0.10-mm TCP-position error and 0.035-N RMS force noise leave aggregate compound pass close to nominal, whereas a positive force bias sharply worsens tracking in B0 and S1. Fifty milliseconds of observation delay reduces both completion and sampled-limit safety, and a 5-degree registered-normal error primarily disrupts task completion. The opposite-signed force bias sometimes improves the thresholded tracking count while increasing force-tail exposure, so calibration error cannot be summarized by a single success rate. These isolated interventions locate deployment sensitivities without representing a joint hardware-error distribution.

The cross-engine results separate portability from task equivalence. The revised MuJoCo port meets the held-out multilevel normal-cycle criteria and matches the bidirectional motion’s force history, onset, and peak, but its 0.833-mm tangential-position RMSE exceeds the 0.5-mm tolerance. Zero-shot deployment is consequently interpreted as a stress test. All 30 flat and inclined evaluations pass, whereas all 15 cylindrical evaluations fail with sampled-limit exceedances. The surface-specific split strengthens the evidence for the simpler geometries, but it does not establish engine-invariant performance. Curved contact remains sensitive to the policy, geometry representation, and solver dynamics together.

TABLE 6. Zero-shot cross-engine stress results on matched checkpoint–block–target keys. Counts are out of 15 evaluations per block.

Engine	Block	Task	Tracking	Sampled safety	Compound	Max. peak (N)
PhysX	Flat reference	15	11	15	11	13.09
MuJoCo	Flat reference	15	15	15	15	13.32
PhysX	10° incline	15	7	15	7	12.30
MuJoCo	10° incline	15	15	15	15	13.20
PhysX	Cylinder, $r = 1.2$ m	12	6	13	6	17.90
MuJoCo	Cylinder, $r = 1.2$ m	0	0	0	0	20.24

The repeated cylinder boundary also identifies a distinct method problem. Prior curved-surface wiping and polishing systems couple local-normal estimation, tool-posture adaptation, and force feedback [10], [11]. Our policy receives a registered local frame but commands translation only; neither its latent force predictor nor its adaptive MPPI objective can rotate the tool to follow a changing normal. A curvature-specific extension should therefore add a bounded posture channel and compare registered, force-derived, and learned local-normal representations under a radius sweep. Retuning the present scalar force costs would not isolate this missing capability. We leave that extension outside the frozen study rather than adapting the method to the observed cylinder blocks.

The numerical audit applies unchanged. Task completion, force tracking, contact continuity, and sampled force tails have different evidence requirements. The final method comparison consequently reports them separately as well as jointly. Peak force is an empirical statistic under the recorded solver configuration. The adaptive planner’s mean peak increases by 0.247 N and both planners record two samples above 15 N, so the measured tracking benefit is not interpreted as a safety improvement. MRE and NRMSE describe regulation on the observed contact samples.

All experiments use rigid-contact simulation and a registered nominal task frame. The observations are causal but do not reproduce a physical camera, force-sensor bandwidth, calibration drift, deformable contamination, or robot communication stack. The factor-separated design improves causal interpretation within simulation. The cross-engine stress test broadens that evidence without replacing hardware validation.

XI. Conclusion

This work introduced RA-RMPPI, a force-conditioned TD-MPC2 controller that adapts both its planning objective and its computation to causal contact regimes. Matched component studies separated objective adaptation from compute adaptation, while behaviour-cloning, router, and risk-representation ablations identified the smallest supported formulation. In the factor-separated simulation study, RA-RMPPI raised compound pass from 85/135 to 100/135, reduced MRE and NRMSE, and used fewer planning samples and less host planning time than the fixed planner built from the same checkpoints.

The improvement is specific to force regulation and computation. Task completion remained 132/135 versus 133/135,

both planners recorded two native samples above 15 N, and the adaptive planner had a higher mean peak force. The cylindrical 12 N condition therefore remains a concrete boundary rather than an exception hidden by aggregate results. A 420-evaluation deployment-gap panel further identified signed force bias, 50-ms delay, and registered-normal error as distinct sensitivities. In a zero-shot MuJoCo stress test, all 30 flat and inclined evaluations achieved the compound outcome, while all 15 weak-curvature evaluations failed; the port’s remaining tangential calibration mismatch precludes a matched-solver claim. Within the evaluated simulation setting, the study shows that continuous regime adaptation can make direct learned wiping more accurate and less computationally demanding, while reliable force-tail control under curvature requires a surface-following posture or learned-constraint extension beyond the present translation-only action interface, followed by physical validation.

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